HEMORA - Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing

25-26J-514

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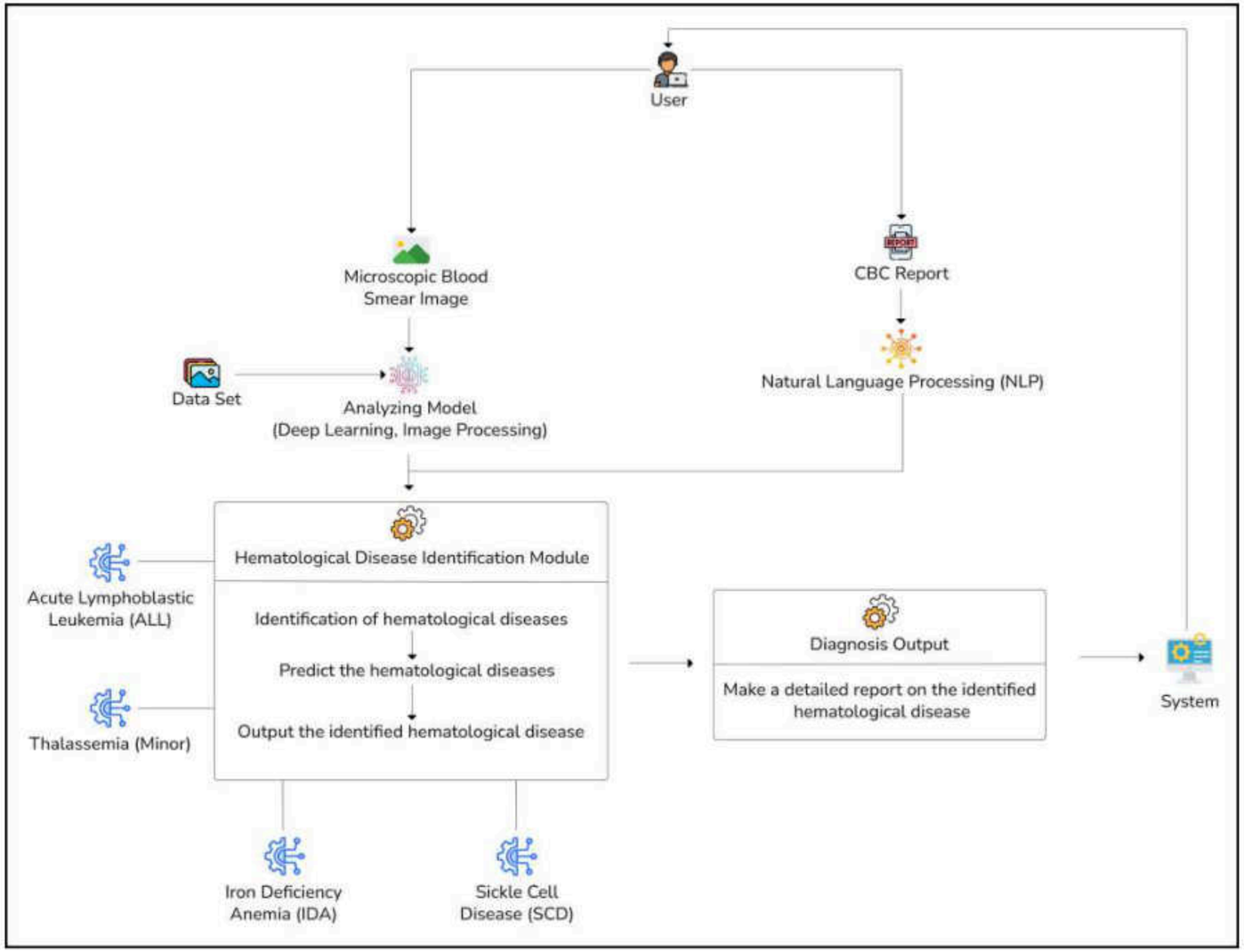
Introduction Video

Overall System Diagram



Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

Specialization: Software Engineering

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IT22371218 | Aluthge A D M S | 25-26J-514

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

Problem Definition

- Acute Lymphoblastic Leukemia (ALL) diagnosis in Sri Lanka has 3-7 day delays in test results due to and shortage of specialist hematologists and limited laboratory facilities
- Traditional manual microscopic examination is error-prone and relies on doctor's subjective assessment

Proven Gap

- Current research approaches ALL detection through isolated methodologies - either blood smear image analysis OR CBC parameter analysis but lacks integration of both modalities
- No published work on integrated hybrid systems for ALL detection tailored to Sri Lankan patient populations, with most international models trained on Western datasets

Existing Systems	Proposed Solution
Single-modal approaches (image-only or CBC-only) lack comprehensive diagnostic capability	Hybrid CNN-YOLO + CBC parameter fusion for comprehensive multi-modal analysis
Trained on Western datasets	Sri Lankan population-specific training data
No real-time clinical decision support tools available	Web-based real-time diagnostic platform with confidence scoring and clinical decision support

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

Key Domains of Knowledge

- **Software Engineering:**
  - System design
  - Hybrid architecture
  - Web-based deployment
- **Deep Learning & Machine Learning:**
  - CNN architectures for medical imaging
  - YOLO for lymphoblast detection
  - Image preprocessing and segmentation
  - CBC parameter classification
  - Decision-level fusion techniques
  - Multi-modal data integration
- **Medical Informatics:**
  - Hematological disease diagnosis
  - Clinical relevance in Sri Lankan context

Specific Latest Technologies Utilized

- **Deep Learning Models:** A hybrid framework will combine the strengths of CNN and YOLO architectures for comprehensive lymphoblast analysis.
- **CNN (Convolutional Neural Network):** Used for high-accuracy morphological feature extraction and lymphoblast classification from blood smear images.
- **YOLO (You Only Look Once):** Utilized for real-time detection and localization of lymphoblasts within peripheral blood smear images.
- **Machine Learning (Random Forest, SVM, Decision Trees):** Analyzes CBC parameters to support diagnosis.
- **Decision-Level Fusion:** Integrates image and CBC results for improved sensitivity and specificity.
- **Vision Transformers (ViTs) / ResNet:** Extracts complex morphological features for precise classification.
- **U-Net / Watershed Segmentation:** Segments WBCs to isolate lymphoblasts and enhance morphology clarity.

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

Evaluation / Success Measurement

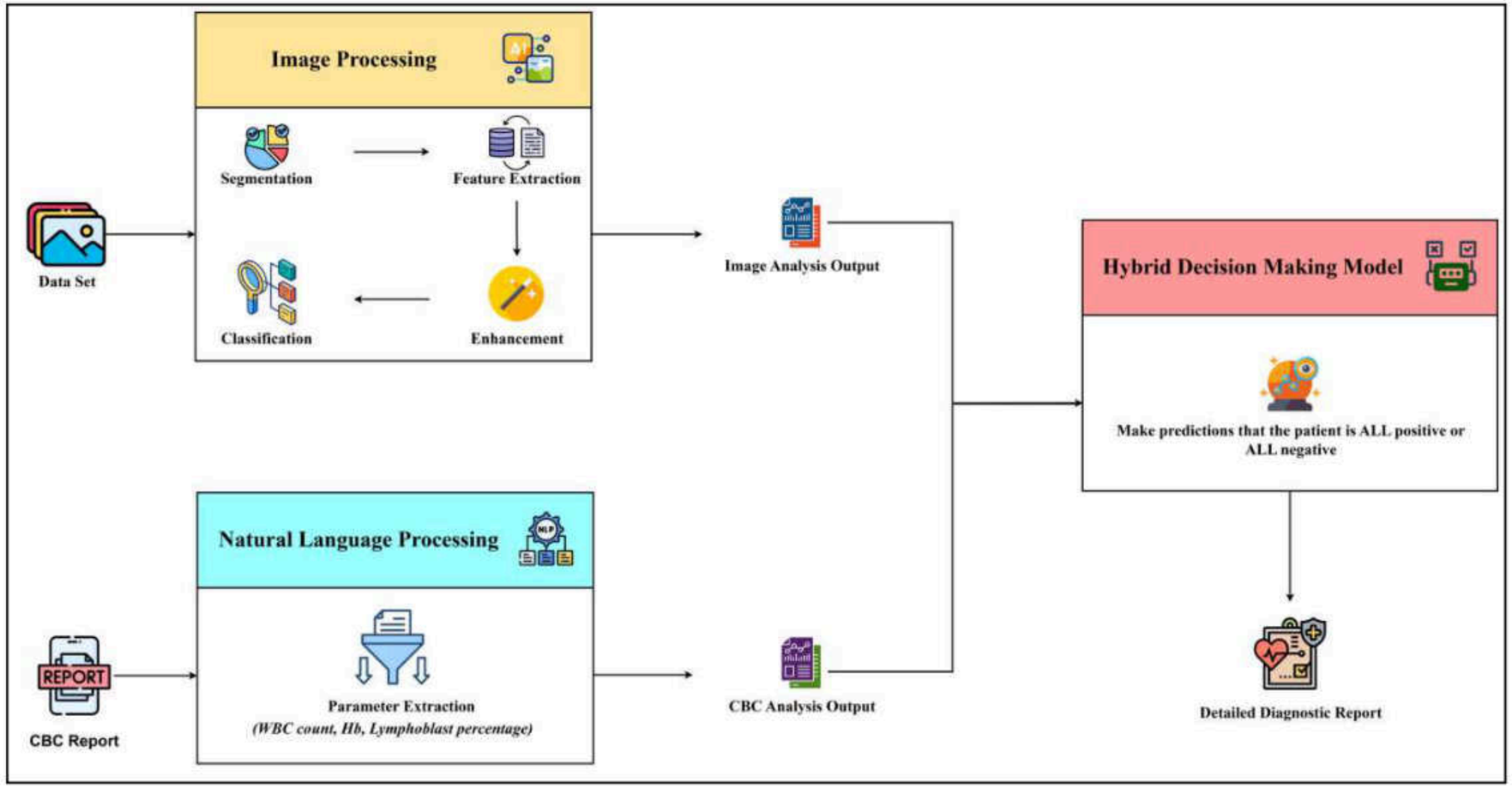
- The hybrid model will be trained and evaluated on both blood smear images and CBC parameters using a locally collected dataset.
- **Target Accuracy:** >90% accuracy
- **Performance metrics:**
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC
- **Validation:** Testing with real clinical data.

Data Availability and Ethical Clearance

- **Data Sources:** Sri Lankan hospitals, diagnostic labs, open datasets.
- **Data Needed:** CBC parameters (WBC count, Hb, lymphoblast %) + PBS images (large nuclei, high N:C ratio, irregular shapes).
- **Ethical Approval:** University Research Ethics Committee + Ministry of Health clearance.
- **Data Privacy:** Anonymized patient data, encrypted and securely stored.

Detection and Identification of Acute Lymphoblastic Leukemia (ALL)

System Overwiew



Real-world Scenario

- The solution will be implemented as a user-friendly, **web-based interface** that is easily accessible to healthcare professionals.
- Laboratory technician uploads **Blood Smear Image & CBC Report.**
- System analyzes using hybrid model
- System automatically generates **Diagnosis Report.**
- This will support doctors in early detection, reduces misdiagnosis.

Detection and Identification of Beta-Thalassemia (Minor)

Specialization: Software Engineering

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Detection and Identification of Beta-Thalassemia (Minor)

Problem Definition

- Current diagnosis relies on manual microscopy, which are costly, time-consuming, and not widely accessible, while no Sri Lanka–specific AI diagnostic system currently exists.
- **Misdiagnosis risk:** Beta-Thalassemia Minor vs. Iron Deficiency Anemia (IDA) due to overlapping CBC indices.

Proven Gap

- There is no hybrid AI model that combines CBC data with blood smear images, and most existing systems are trained on foreign datasets, making them unsuitable for Sri Lankan patients.
- Furthermore, current methods struggle to differentiate Beta-Thalassemia Minor from IDA and lack a web-based platform for real-time clinical deployment.

Existing Systems	Proposed Solution
CBC-based ML Models : Cannot differentiate Thalassemia vs IDA	**Hybrid model (CBC + Smear images)**
Blood Smear DL Models : Ignore numerical indices	**CNN for morphology + CBC integration**
Foreign Dataset Models : Not tuned for Sri Lankan variations	**Train on Sri Lankan patient data**

Detection and Identification of Beta-Thalassemia (Minor)

Key Domains of Knowledge

- **Software Engineering:**
  - System design
  - Hybrid architecture
  - Web-based deployment.
- **Machine Learning & Deep Learning:**
  - RBC morphology classification
  - CBC parameter modeling.
  - Image preprocessing and segmentation
- **Medical Informatics:**
  - Clinical relevance in Sri Lankan hematology context.
  - Hematological disease diagnosis

Specific Latest Technologies Utilized

- **Hybrid Deep Learning Models:**
Combines CNN architectures with feature-level fusion for higher diagnostic accuracy.

- **U-Net Segmentation:**
Extracts and isolates red blood cells to improve clarity of morphological features.

- **ResNet / EfficientNet:**
Classifies RBC morphologies such as microcytosis, hypochromia, and target cells.

- **Machine Learning Models (Random Forest, SVM):**
Analyzes CBC indices (MCV, MCH, RBC count) to support hybrid decision-making.

Detection and Identification of Beta-Thalassemia (Minor)

Evaluation / Success Mesurement

- The hybrid model will be trained and evaluated on both blood smear images and CBC parameters using a locally collected dataset.

- **Performance metrics:**
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC

- **Target Accuracy:** >90% accuracy

- **Validation:**
  - Testing with real clinical data.

Data Availability and Ethical Clearance

- **Data Sources:**

Sri Lankan hospitals, diagnostic labs, open datasets.

- **Data Needed:**

CBC parameters (MCV, MCH, RBC, Hb) + PBS images (microcytosis, hypochromia, target cells).

- **Ethical Approval:**

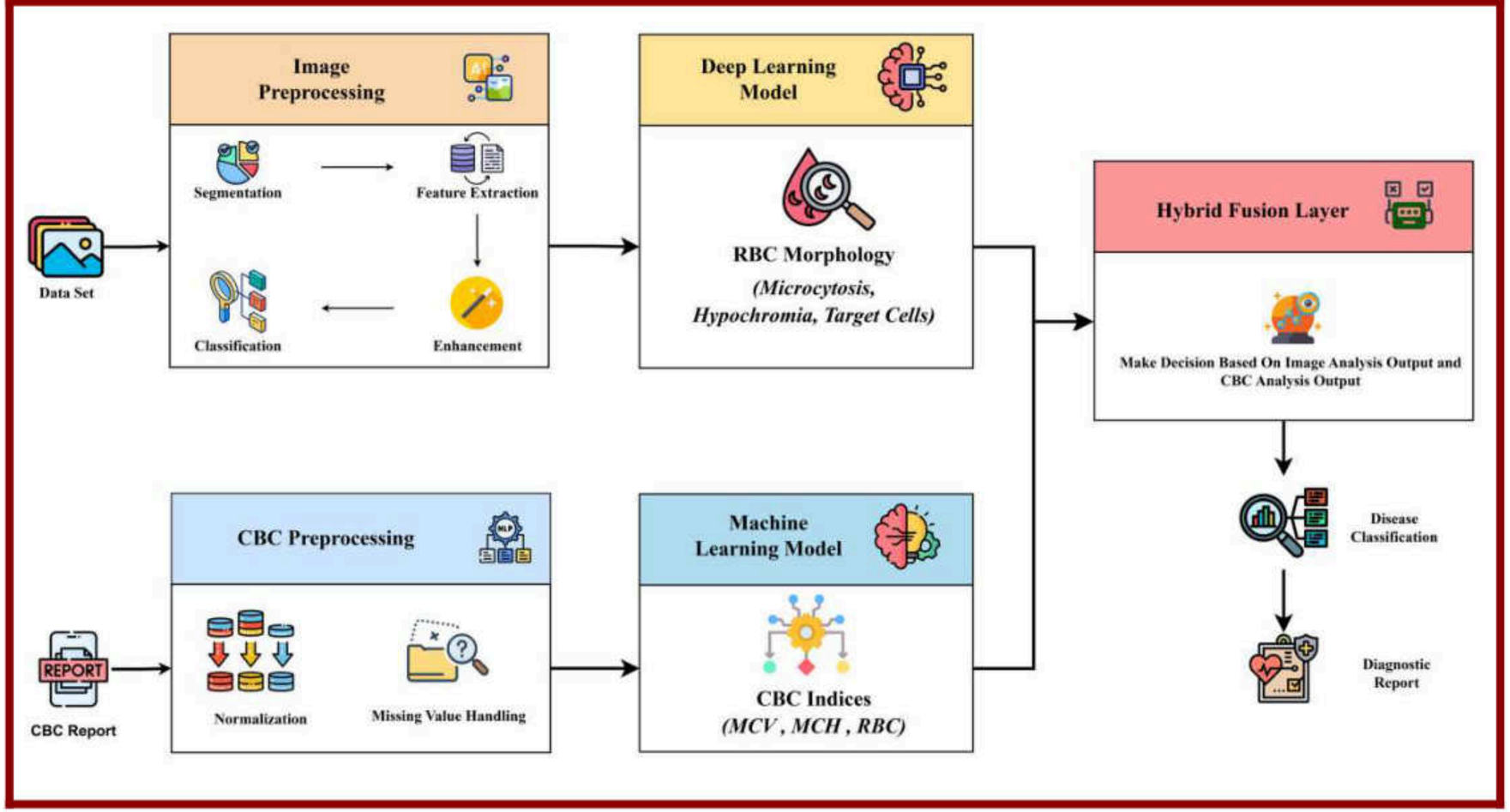
University Research Ethics Committee + Ministry of Health clearance.

- **Data Privacy:**

Anonymized patient records, secure storage.

Detection and Identification of Beta-Thalassemia (Minor)

System Overview Diagram



Real-world scenario

- Laboratory technician uploads **Blood Smear Image & CBC Report.**
- System automatically generates **Diagnosis Report.**
- Supports doctors in early detection, reduces misdiagnosis.
- **Web-based System:** Scalable for use in Sri Lankan hospitals, diagnostic labs, and rural health centers.

Detection and Identification of Sickle Cell Disease (SCD)

Specialization: Software Engineering

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Detection and identification of Sickle Cell Disease (SCD)

Problem Definition

- Current process is manual microscopic examination, which prone to human error and inter-observer variability, will eventually delay the diagnosis process.
- Significant shortage of trained hematologists and laboratory technicians, especially in rural areas, further delays diagnosis and treatment.

Proven Gap

- There is no hybrid AI model that combines CBC data with blood smear images, and most existing systems are trained on foreign datasets, making them unsuitable for Sri Lankan patients.
- Furthermore, current methods struggle to differentiate Beta-Thalassemia Minor from IDA and lack a web-based platform for real-time clinical deployment.

Existing Systems	Proposed Solution
Dataset structure : Not based on CBC data, only images	**CBC Reports + Blood Smear images**
Deep Learning Models : Single R-CNN model on cell identification	**Hybrid model (R-CNN + YOLO)**
Less Dataset Models : 150 training + 20 validation images	**More than 500+ images will be used**

Detection and identification of Sickle Cell Disease (SCD)

Key Domains of Knowledge

- **Software Engineering:**
  - System design
  - Hybrid architecture
  - Web-based deployment.
- **Machine Learning & Deep Learning:**
  - RBC morphology classification
  - CBC parameter modeling.
  - Image preprocessing and segmentation
- **Medical Informatics:**
  - Clinical relevance in Sri Lankan hematology context.
  - Hematological disease diagnosis

Specific Latest Technologies Utilized

- **Hybrid Deep Learning Models:** Combines CNN architectures with feature-level fusion for higher diagnostic accuracy.
- **R-CNN (Region-based Convolutional Neural Network):** Used for high-accuracy object detection of sickle-shaped cells.
- **YOLO (You Only Look Once):** Utilized for real-time segmentation and analysis of blood cells.
- **Machine Learning Models (Random Forest, SVM):** Analyzes CBC indices (MCV, MCH, RBC count) to support hybrid decision-making.

Detection and identification of Sickle Cell Disease (SCD)

Evaluation / Success Mesurement

- The hybrid model will be trained and evaluated on both blood smear images and CBC parameters using a locally collected dataset.

- **Performance metrics:**
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC

- **Target Accuracy:** >90% accuracy

- **Validation:**
  - Testing with real clinical data.

Data Availability and Ethical Clearance

- **Data Sources:**

Sri Lankan hospitals (government and private), diagnostic labs, open datasets.

- **Data Needed:**

Peripheral Blood Smear images (microcytosis, hypochromia, target cells) + CBC parameters (MCV, MCH, RBC, Hb).

- **Ethical Approval:**

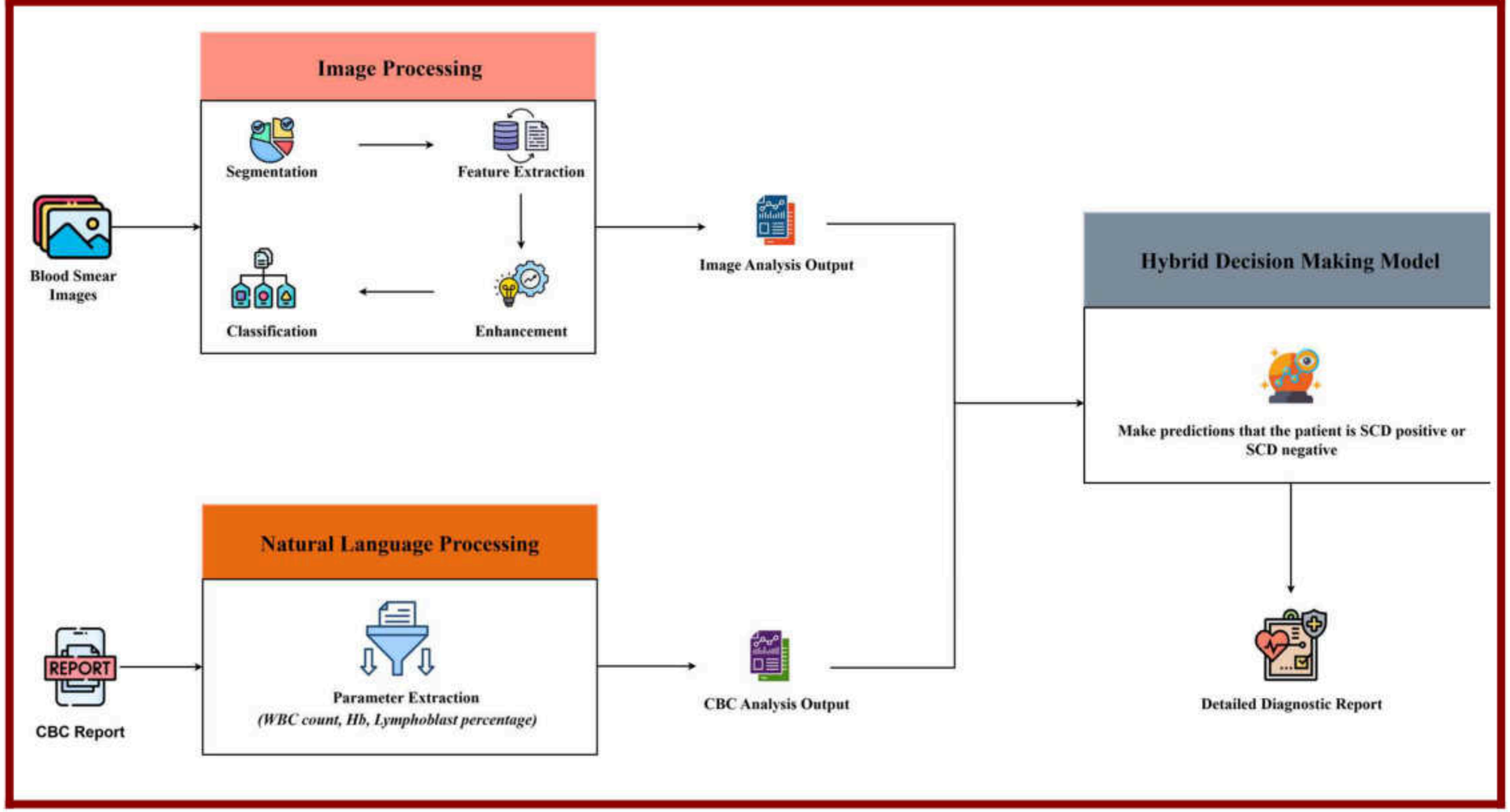
University Research Ethics Committee + Ministry of Health clearance + Approval of ethical committees in hospitals.

- **Data Privacy:**

Anonymized patient records, secure storage.

Detection and identification of Sickle Cell Disease (SCD)

System Overview Diagram



Real-world scenario

- Laboratory technician uploads **Blood Smear Image & CBC Report.**
- System automatically generates **Diagnosis Report.**
- Supports doctors in early detection, reduces misdiagnosis.
- **Web-based System:** Scalable for use in Sri Lankan hospitals, diagnostic labs, and rural health centers.

Detection and Identification of Iron Deficiency Anemia (IDA)

Specialization: Software Engineering

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a (IDA)

dependent on specialists.

validated hybrid model that
ers with blood smear image
of IDA.
w, manual, and depend on
haematologists, causing diagnostic delays.

- **Solution: A system combining CBC parameters with blood smear image features with Sri Lanka specific dataset.**

CBC only or PBS only	Hybrid CBC + PBS model for IDA.
Trained on foreign datasets.	Validated on Sri Lankan data.
Manual, slow, specialist-dependent.	Automated, faster, less specialist-dependent.

Detection and Identification of Iron Deficiency Anemia (IDA)

Key Domains of Knowledge

- **Software Engineering:**
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Specific Latest Technologies Utilized

- **Hybrid Deep Learning Models:**
  Combines CNN architectures with feature-level fusion for higher diagnostic accuracy.
- **U-Net Segmentation:**
  Extracts and isolates red blood cells to improve clarity of morphological features.
- **Vision Transformers(ViTs) / ResNet :**
  Extract morphological features such as microcytosis, hypochromia, anisocytosis and poikilocytosis.
- **Machine Learning Models (Random Forest, SVM):**
  Analyzes CBC indices (Hb, MCV, MCH, RDW) to support hybrid decision making.

Detection and Identification of Iron Deficiency Anemia (IDA)

Evaluation / Success Mesurement

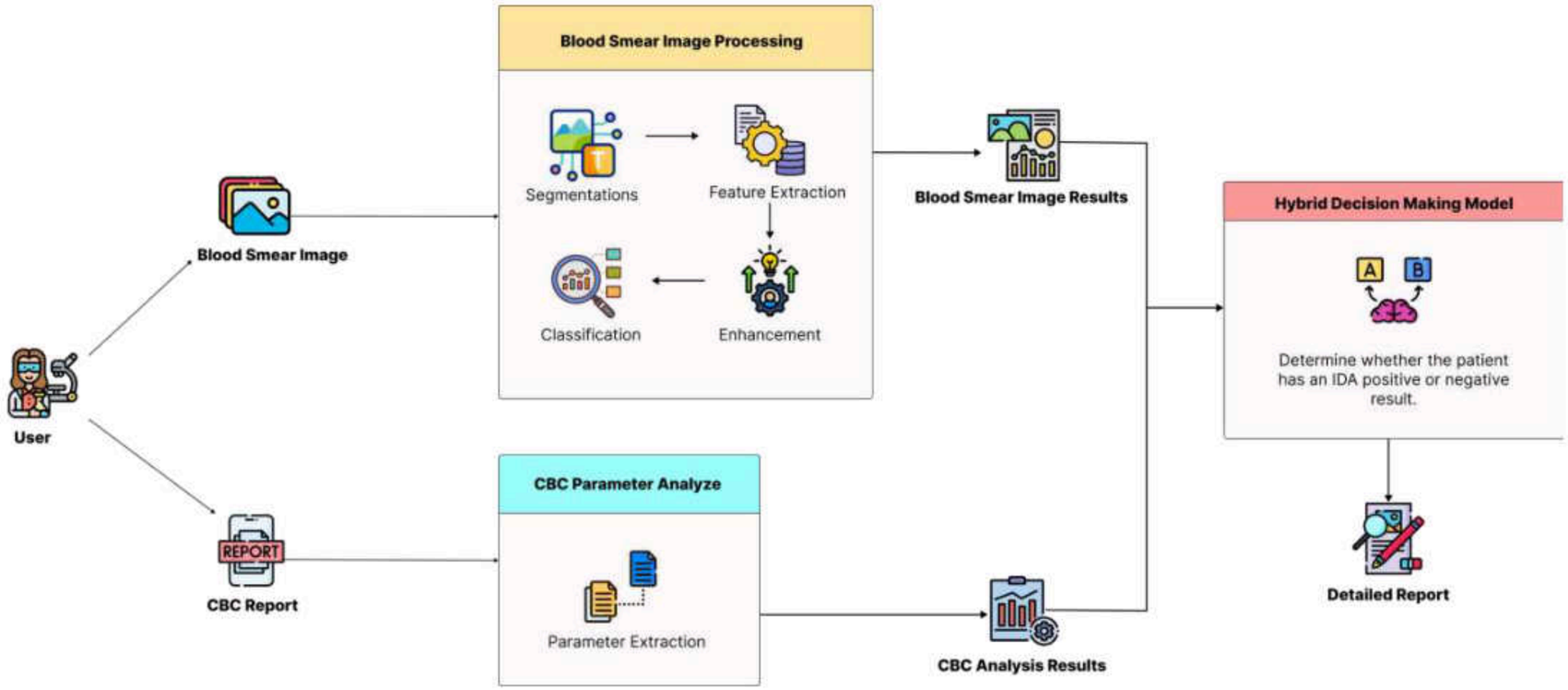
- The model will be trained and validated on a Sri Lanka–specific dataset of CBC reports and PBS images.
- **Performance metrics:**
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC
- **Target Accuracy:** >90% accuracy
- **Validation:**
  - Compare model predictions with ground-truth diagnoses given by haematologists.

Data Availability and Ethical Clearance

- **Data Sources:**
  - CBC reports and PBS images collected from Sri Lankan hospitals and diagnostic labs.
- **Data Needed:**
  - CBC parameters (Hb, MCV, MCH, RDW) + PBS images (microcytosis, hypochromia, anisocytosis, poikilocytosis).
- **Ethical Clearance:**
  - Formal approval obtained from the Sri Lankan Ministry of Health / Ethics Review Committees.
- **Data Privacy:**
  - Patients' personal details anonymised; all data securely stored.

Detection and Identification of Iron Deficiency Anemia (IDA)

System Overview Diagram



Real-world scenario

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- System automatically generates **Diagnosis Report.**
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- **Web-based System:** Scalable for use in Sri Lankan hospitals, diagnostic labs, and rural health centers.

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Ability of Commercialization

- Deployable in hospitals, labs, rural healthcare centers
- Web-based platform ensures low entry barrier, integration with existing workflows

Testing

- **Clinical Testing:** Pilot trials in Sri Lankan hospitals with real patient data.
- Ensures accuracy, reliability, and clinical applicability before commercialization.

Investment Cost

- Material / Infrastructure: ~ LKR.50,000
  - Cloud servers, secure hosting, storage, testing equipment
- Others: ~ LKR.10,000
- Total Estimate Investment Cost ~ LKR.60,000

Cost Recovery

- Subscription for hospitals/labs (e.g., LKR 10,000 – 15,000 / month)
- Pay-per-use for clinics (e.g., LKR 200 – 300 / test)
- Possible government/NGO funding for scaling

Value Proposition

- Fast, accurate & low-cost diagnostics (Leukemia, Thalassemia, IDA, SCD)
- Reduces dependency on specialists & human errors
- Enables early detection → better survival rates

Intellectual Property Rights

- Patentable AI algorithms & hybrid diagnostic framework
- Proprietary dataset/preprocessing pipeline
- Trademark potential for Hemora

Target Market & Customer Profile

- **Customers:** Hospitals, diagnostic labs, NGOs, rural clinics
- **Users:** Hematologists, lab technicians, physicians